Answer Key

Physics			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
1	(3)	Thermal Conductivity	Thermal Properties of Matter
2	(4)	Thermodynamic Process	Kinetic Theory of Gases
3	(3)	Electric Dipole	Electrostatics
4	(3)	Photoelectric Effect	Dual Nature of Radiation and Matter
5	(1)	Electric Flux	Electrostatics
6	(2)	Dimensions	Units and Measurements
7	(4)	Focal Length of Lens	Ray Optics
8	(2)	Representation of EMW	Electromagnetic Waves
9	(2)	Refraction at Curved Surface	Ray Optics
10	(1)	Spring	Oscillations
11	(4)	Elastic Collision	Work, Energy and Power
12	(4)	Power	Work, Energy and Power
13	(2)	Focal Length of Lens	Ray Optics
14	(3)	Charging and Discharging of Capacitor	Electrostatics
15	(3)	Angular Momentum	System of Particles and Rotational Motion
16	(4)	Dimensions	Units and Measurements
17	(2)	Unit of Magnetic Quantities	Moving Charges and Magnetism
18	(1)	Logic Gates	Semiconductor Electronics
19	(1)	Newton's Law of Cooling	Thermal Properties of Matter
20	(1)	Hydrogen Spectrum	Atoms and Nuclei
21	[8]	Solenoid	Moving Charges and Magnetism
22	[1320]	Parallel Plate Capacitor	Electrostatics
23	[7700]	Errors in Measurements	Units and Measurements
24	[8]	Velocity of Satellites, Angular Momentum	Gravitation
25	[3]	Acceleration	Motion in 1-D

Chemistry			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
26	(2)	Magnetic Moment	Coordination Chemistry
27	(2)	Energy Diagram	Structure of Atom
28	(2)	Chromatography	Practical Organic Chemistry
39	(1)	Amino Acid Classification	Biomolecules
30	(4)	Nucleophillic Substitution Reaction	Halo Alkane
31	(1)	Le Chatelier Principle	Equilibrium
32	(2)	Depression in Freezing Point	Liquid Solution

33	(3)	Directing Nature of Group	Aromatic Hydrocarbon
34	(1)	Redox Titration	Redox Reaction
35	(1)	Types of Batteries	Electrochemistry
36	(1)	Types of Product During Electrolysis	Electrochemistry
37	(1)	Types of Complexes	Coordination Chemistry
38	(1)	Number of Sigma and Pi Bond	Chemical Bonding
39	(4)	Calculation of Enthalpy	Thermodynamics
40	(2)	Heisenberg Uncertainty Principle	Structure of Atom
41	(1)	Mixed Reaction of Organic Chemistry	Amines
42	(4)	First Order Reaction	Chemical Kinetics
43	(1)	Types of Oxides	Periodic Classification
44	(1)	Group 15	p Block
45	(2)	Preparation of Phenol	Phenol
46	[0]	Magnetic Moment	Coordination Chemistry
47	[2]	Baeyer's Test	Hydrocarbon
48	[60]	Percentage Yield	Some Basic Concept of Chemistry
49	[275]	Percentage Composition of Sulphur	Practical Organic Chemistry
50	[12]	Calculation of Number of Electrons	Chemical Bonding

Mathematics

Q. No.	Answer Key	Topic's Name	Chapter's Name
51	(3)	Roots of Quadratic Equation	Quadratic Equation
52	(3)	Trigonometric Identities	Trigonometry
53	(4)	Area between 2 Curves	Area under the Curve
54	(3)	Domain	Function
55	(1)	Differentiability	Continuity and Differentiability
56	(3)	Distance	Vector - Three Dimensional
57	(2)	Properties	Sequence and Series
58	(4)	Point and Line	Straight Line
59	(3)	Chord	Ellipse
60	(3)	Rank	Permutation and Combination
61	(1)	Solutions of system of Linear Equations	Determinant
62	(4)	Positive Integral Power of Matrix	Matrices
63	(1)	Point and Line	Three Dimensional
64	(2)	Chord	Circle
65	(2)	Probability Distribution	Probability
66	(4)	Bayes Theorem	Probability

67	(1)	Remainder Theorem	Binomial Theorem
68	(1)	Leibnitz Rule	Definite Integration
69	(3)	Product of Vectors	Vectors -Three Dimensional
70	(1)	Linear Differential Equation	Differential Equation
71	[12]	Properties of Definite Integration	Definite Integration
72	[64]	Special Limits	Limits
73	[11132]	AP	Sequence and Series
74	[10]	Determinant	Determinant
75	[1328]	Circle and other Curve	Conic Section

Answers with Explanation

Physics

1. Option (3) is correct.

As per Seebeck effect, electrical voltage generates when two ends of a metal is at different temperature. To maintain temperature difference, material must have low thermal conductivity otherwise heat will flow easily reducing temperature difference. Also material should have high electrical conductivity, which will allow easy movement of charges resulting into electrical energy.

2. Option (3) is correct.

Isothermal process $PV = \text{const}$, $\frac{dP}{dV} = \frac{-P}{V}$

Adiabetic process $PV^{\gamma} = \text{const}$, $\frac{dP}{dV} = \frac{-\gamma P}{V}$

Slope of adiabetic = $\gamma \times$ slope of isothermal.

Since, $\gamma > 1$, Slope of adiabetic process is steeper than that of isothermal process.

$$\left.\frac{\Delta P}{\Delta V}\right|_{adiab} > \left.\frac{\Delta P}{\Delta V}\right|_{isoth}$$

For a given change in pressure (ΔP),

$$\Delta V\big|_{isoth} > \Delta V\big|_{adiab}$$

i.e., volume falls off more rapidly in the isothermal process than in the adiabetic process.

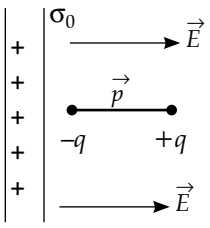
3. Option (3) is correct.

Electric field due to an infinite sheet is given by $E = \frac{\sigma_0}{2\varepsilon_0}$.

Since, $|\vec{E}|$ is constant, the net force as well as the torque on the dipole in zero.

$|\vec{E}|$ and $|\vec{p}|$ are parallel vectors,

$U = -pE\cos 0° = -pE$,



Potential energy of the dipole in minimum (stable equilibrium).

4. Option (3) is correct.

In photoelectric effect,

$eV_0 = KE_{\max}$

$V_0 = \frac{1}{e} \times KE_{\max}$

As frequency of the incident radiation increases, $KE_{\max}$ of the emitted electrons increases and V_0 (stopping potential) also increases.

5. Option (1) is correct.

From Gauss's theorem,

$$\phi_E = \frac{q_{encl}}{\varepsilon_0}$$

$$q_{\text{enclosed}} = \varepsilon_0 \phi_E$$

$$= -8.85 \times 10^{-12} \times 2 \times 10^4$$

$$= -17.7 \times 10^{-8}\text{C}.$$

6. Option (2) is correct.

No heat is absorbed in an adiabatic process AB.

For isothermal process BC, $\Delta U = 0$

$$\Delta Q = \Delta W = nRT \ln\left(\frac{V_f}{V_i}\right)$$

$$\frac{\Delta Q}{n} = R(450)\ln\left(\frac{V_C}{V_B}\right)$$

$$= 450R\ln\left(\frac{4}{3}\right)$$

$$= 450R(\ln 4 - \ln 3).$$

7. Option (4) is correct.

For a thin lens $\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

For L_1 : $\frac{1}{f_1} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{-R}\right) = 2\left(\frac{\mu - 1}{R}\right)$

For L_3 : $\frac{1}{f_2} = (\mu - 1)\left(\frac{1}{R} - \frac{1}{\infty}\right) = \left(\frac{\mu - 1}{R}\right)$

$\therefore \frac{f_1}{f_2} = 1:2$

8. Option (2) is correct.

Direction of $\vec{E}, \vec{B}$ and $\vec{c}$ in an electromagnetic wave are related as $\vec{E} \times \vec{B} = \vec{c}$ (wave moves in $+x$-direction) $\hat{j} \times \hat{k} = \hat{i}$.

9. Option (2) is correct.

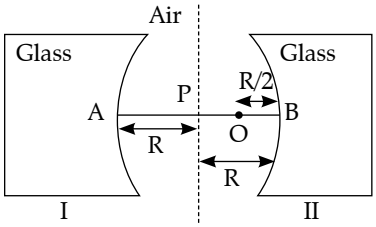


Image formation due to a single refracting surface is given by

$$\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

For surface I, $\frac{1.5}{v_1}-\frac{1}{-3R/2}=\frac{1.5-1}{-R}$

$$\frac{1.5}{v_1}+\frac{2}{3R}=\frac{-0.5}{R}$$

$$\frac{1.5}{v_1}=-\frac{1}{2R}-\frac{2}{3R}=\frac{-7}{6R}$$

$$v_1=-\frac{9R}{7}=-1.286R$$

For surface II, $\frac{1.5}{v_2}-\frac{1}{-R/2}=\frac{1.5-1}{-R}$

$$\frac{1.5}{v_2}=\frac{-2}{R}-\frac{0.5}{R}=\frac{-2}{R}-\frac{1}{2R}$$

$$\frac{1.5}{v_2}=\frac{-5}{2R}$$

$v_2 = -0.6R$

The distance between the two images is

$d = 2R - (v_1 + v_2) = 0.114$ R.

10. Option (1) is correct.

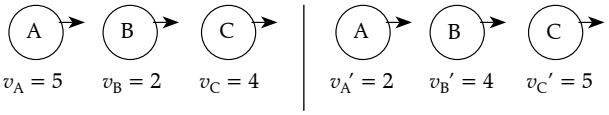
Maximum velocity is SHM = Aω

where, $\omega=\sqrt{\frac{k}{m}}$

$$\frac{v_{max}A}{v_{max}B}=\frac{\omega_1}{\omega_2}=\sqrt{\frac{k_1}{k_2}}$$

11. Option (4) is correct.

In an elastic collision between two equal masses, colliding bodies exchange velocities.



Before After

Hence, the assertion is false but the reason is true.

12. Option (4) is correct.

Given, $\frac{dm}{dt}=\beta\sqrt{v}$ where β = constant.

Power $P = Fv = (ma)v = m\left[v\frac{dm}{dt}\right]v = mv^2\left(\beta\sqrt{v}\right)$

$P \propto v^{5/2}$

$P^2 \propto v^5$.

13. Option (2) is correct.

Focal length of the lens in air (f_a) and in water (f_w) are related as follows:

$$f_w = f_a\left(\frac{\mu-1}{\frac{\mu}{\mu_w}-1}\right)$$

Where, μ = refractive index of glass and

μ_w = refractive index of water.

$$f_w = 24\times\frac{(1.5-1)}{\left(\frac{1.5}{1.33}-1\right)}=\frac{24\times0.5\times1.33}{0.17}$$

$= \sim 94$ cm. (nearest option is 96 option)

14. Option (3) is correct.

At equilibrium, both acquire a common potential:

$$V'=\frac{C_1V_1+C_2V_2}{C_1+C_2}=\frac{(6\times5)+(12\times0)}{6+12}=\frac{30}{18}=\frac{5}{3}\text{V}$$

$$\therefore\ q_1 = C_1V' = 6\times\frac{5}{3}=10\,\mu C$$

$$q_2 = C_2V' = 12\times\frac{5}{3}=20\,\mu C.$$

15. Option (3) is correct.

Since, all masses are moving under mutual gravitational force, $\vec{\tau}_{sys}=\vec{0}$, the angular momentum of the system is conserved.

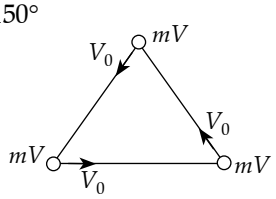
Hence, the angular momentum of the system at the point of collision is the same as initial angular momentum of the system.

i.e.

$$= 3\times mV_0a\left(\frac{2}{\sqrt{3}}\right)\sin150°$$

$$= \sqrt{3}mV_0a\cos60°$$

$$= \frac{\sqrt{3}}{2}mV_0a$$



16. Option (4) is correct.

[Young's Modulus] = $\left[\frac{F}{A}\right]=\left[M^0L^{-1}T^{-2}\right]$

[Torque] = F.r = $[M^1L^1T^{-2}]$ [L^1] = $\left[M^1L^2T^{-2}\right]$

[Coefficient of viscosity] = $\left[M^1L^{-1}T^{-1}\right]$

[Gravitational constant] = $\left[M^{-1}L^3T^{-2}\right]$

17. Option (2) is correct.

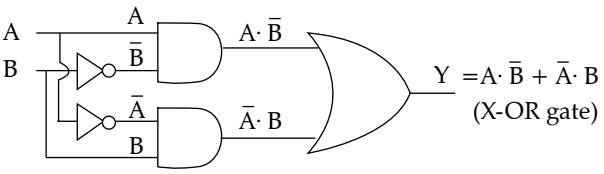
Magnetic induction → gauss

Magnetic intensity → amp/m

Magnetic flux → weber

Magnetic moment → amp m^2

18. Option (1) is correct.



The truth table for X-OR gate is:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

19. Option (1) is correct.

Applying Newton's law of cooling

$\frac{\Delta T}{\Delta t} = k\left(\overline{T} - T_0\right)$ where $k =$ constant,

$\overline{T}$ = average temperature,

T_0 = room temperature.

$$\frac{90° - 80°}{t} = k\left(\frac{90° + 80°}{2} - 20°\right)$$

$$\frac{10^0}{t} = k(85° - 20°) = k\,65° \qquad \ldots(i)$$

$$\frac{80° - 60°}{t'} = k\left(\frac{80° + 60°}{2} - 20°\right)$$

$$\frac{20°}{t'} = k(70° - 20°) = 50°k \qquad \ldots(ii)$$

Dividing equation (i) by (ii):

$$\frac{1}{2} \times \frac{t'}{t} = \frac{65}{50}$$

$$t' = \frac{13}{5}t.$$

20. Option (1) is correct.

The number of spectral lines (N) emitted when an atom de-excites is given by $\frac{n(n-1)}{2}$.

Here, $n = 4$.

$$\therefore\ N = \frac{4 \times 3}{2} = 6$$

21. Correct answer is [8].

Magnetic field inside a solenoid $B = \mu_0\left(\frac{N}{l}\right)i$

$$l = \frac{\mu_0 Ni}{B} = \frac{4\pi \times 10^{-7} \times 200 \times 0.29}{2.9 \times 10^{-4}}\text{m}$$

$= 8\pi \times 10^{-2}$ m

$= 8\pi \times 10^{-2} \times 10^2$ cm

$= 8\pi$ cm.

22. Correct answer is [1320].

For a capacitor $q = CV$

$$i = \frac{dq}{dt} = C\frac{dV}{dt}$$

$$\therefore\ C = \frac{i}{(dV/dt)} = \frac{0.15}{7 \times 10^8} = \frac{\varepsilon_0 A}{d}$$

$$\therefore\ d = \varepsilon_0 A \times \frac{7 \times 10^8}{0.15}$$

$$= 9 \times 10^{-12} \times \frac{22}{7} \times 0.01 \times \frac{7 \times 10^8}{0.15} = 1320\ \mu\text{m}.$$

23. Correct answer is [7700].

$$Q = \frac{ab^4}{cd}$$

$$\frac{\Delta Q}{Q} = \frac{\Delta a}{a} + \frac{4\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d}$$

$$\%\frac{\Delta Q}{Q} = \left[\frac{3}{60} + 4\left(\frac{0.1}{20}\right) + \frac{0.2}{40} + \frac{0.1}{50}\right] \times 100 = 7.7$$

$$= \frac{77}{10}$$

$$= \frac{7700}{1000}$$

$$= \frac{x}{1000}$$

$$\therefore\ x = 7700.$$

24. Correct answer is [8].

Angular momentum of a planet about the star L = *mvr*,

where, v= orbital velocity = $\sqrt{\frac{GM}{r}}$

$$|\vec{L}| = m\sqrt{\frac{GM}{r}}\sqrt{r^2} \Rightarrow |\vec{L}| \propto m\sqrt{r}$$

$$\frac{|\vec{L}_B|}{|\vec{L}_A|} = \frac{m_B}{m_A}\sqrt{\frac{r_B}{r_A}} = 4\sqrt{2} \times \sqrt{2} = 8.$$

25. Correct answer is [3].

$a_P = k_1 t$;

where, k_1 = constant

$a_Q = k_2$ = constant

$$a_p = k_1 t = \frac{dv_p}{dt} \Rightarrow v_p = \frac{k_1 t^2}{2} = \frac{ds_p}{dt}$$

$$\Rightarrow s_p = \frac{k_1 t^3}{6}$$

similarily

$$\frac{k_1}{6}t^3 - \frac{1}{2}k_2 t^2 = 0$$

$$a_Q = k_2 = \frac{dv_Q}{dt} \Rightarrow v_Q = k_2 t = \frac{ds_Q}{dt}$$

$$\Rightarrow s_Q = \left(\frac{1}{2}\right)k_2 t^2$$

Both cross each other at $t = 0$ for the first time. Wherever they cross each other, the distance from the origin is same for both.

$s_p = s_Q$

$$\frac{k_1}{6}t^3 = \frac{1}{2}k_2 t^2$$

Thus, obtained a cubic polynomial which has 3 roots. Hence, the maximum no of crossings will be 3.

Chemistry

26. Option (2) is correct.

The oxidation state of Fe in $K_3[Fe(OH)_6]$ and $K_4[Fe(OH)_6]$ complex are +3 and +2 respectively.

In complex $K_3[Fe(OH)_6]$, the electronic configuration of Fe^{+3} is as follows:

Fe ($Z = 26$): $3d^6\ 4s^2$

Fe^{+3}: $3d^5\ 4s^0$

Being a weak field ligand, OH does not pair with the unpaired electron.

$\therefore$ all 5 electrons present in 3d orbital remains unpaired.

For calculation of the spin-only magnetic moment, the following formulae were used:

$\mu = \sqrt{n(n+2)}$ BM, where n = number of unpaired e^-

for $K_3[Fe(OH)_6]$ $n = 5$

$\mu = \sqrt{5(5+2)} = \sqrt{35}$ B.M

μ = 5.92 B.M.

for $K_4[Fe(OH)_6]$ $n = 4$

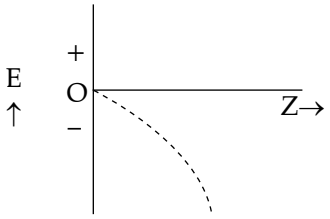
$\mu = \sqrt{4(4+2)} = \sqrt{24}$ B.M.

=4.90 B.M.

27. Option (2) is correct.

From the formula $E_n = -13.6 \times \frac{Z^2}{n^2}$ eV/atom.

Here, as the value of Z increases, the value of E increases. So we obtain the following graph:



28. Option (2) is correct.

Statement I is correct: In partition chromatography, stationary phase is this film of liquid present in the inert support.

Statement II is incorrect: Chromatography paper contains water trapped in it, which acts as the stationary phase, not the material of paper.

29. Option (1) is correct.

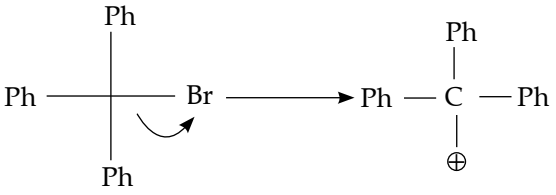
Essential amino acid are those amino acid which do not synthesize in our body and must be obtained through diet.

The essential amino acids are Valine, Lysine and Threonine.

Remember: I Like Lysine very much, please; Isoleucine, Leucine, Lysine, Valine, methionine and pherys alanine. Try to Help Arginine, Tryptophan, threonine, histidine & Arginine

30. Option (4) is correct.

The most stable carbocation in the nucleophilic substitution reaction is a tertiary carbocation which is obtained in option (4).



Here, positive charge in carbon is the resonance stabilised due to the presence of 3 phenyl rings.

31. Option (1) is correct.

On increasing the pressure of the system by twofold at constant temperature the

(A) concentration of both reactant and products increase because with the increase in pressure, volume decreases, which results in an increase in the concentration. So, statement (A) is correct.

(B) Here, the values of $\Delta ng < O$ $\therefore$ On increasing the pressure, equilibrium will shift in the forward direction. So, statement (B) is correct.

The value of equilibrium constant (K) does not change with the concentration; it depends on the temperature only.

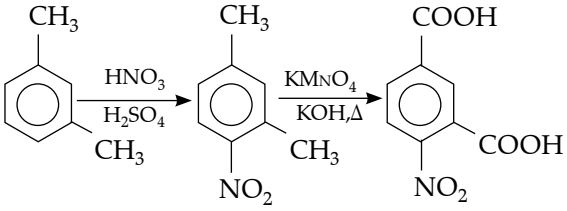
32. Option (2) is correct.

On addition of NaCl, the freezing point of water decreases, which is known as depression in freezing point.

On addition of a non-volatile solute like NaCl, decrease the freezing point due to the decrease in vapour pressure.

So, both statements (I) and (II) are true.

33. Option (3) is correct.



4- Nitro benzene –1, 3 - dicarbaxylic (Major product) Acid

Statement (II) is also correct.

Being e^- donating group $-CH_3$ direct upcoming electrophile towards ortho and para position $\therefore$ It is O/P directing while

$-NO_2$ direct upcoming electrophile towards meta position $\therefore$ it is meta directing

34. Option (1) is correct.

(A) From the reaction it is clear that the equivalent of KI = equivalent of KIO_3.

$M \times V_{(L)}$ = mol × n-factor

$0.1 \times 0.2L$ = mol × 5

$$\text{mol of } KIO_3 = \frac{0.1\frac{\text{mol}}{\text{L}} \times 0.2\text{L}}{5} = 0\cdot004\,\text{mol}$$

Statement (A) is correct.

(C) From the reaction

eq of KI = eq of I_2

$M \times V_{(L)}$ = mol × vf

0.1×0.5 = mol × 1

mol of I_2 = 0.05 mol

Statement (C) is incorrect

(D) The change in the oxidation state of iodine from

$KIO_3 \rightarrow I_2$ is 5.

$$\text{Equivalent weight of } KIO_3 = \left(\frac{\text{mol}\cdot\text{wt}}{5}\right)$$

Statement (D) is correct.

35. Option (1) is correct.

S No.	List-I Applications	List-II Batteries/Cell
A.	Transistors	(iii) Anode – Zn; Cathode – carbon
B.	Hearing aids	(i) Anode – Zn/Hg; Cathode – HgO + C
C.	Invertors	(iv) Anode – Pb; Cathode – Pb/[Pb O_2]
D.	Apollo space ship	(ii) Hydrogen fuel cell

36. Option (1) is correct.

(A) An aqueous solution of $AgNO_3$ using silver electrodes.

Anode: $Ag_{(s)} \rightarrow Ag^+ + e^-$

Cathode: $Ag^+_{(aq)} + e^- \rightarrow Ag_{(s)}$

$$\left[Ag_{(s)}\right]_{\text{Anode}} \rightarrow \left[Ag_{(s)}\right]_{\text{Cathode}}$$

No O_2 gas is evolved.

(B) An aqueous solution of $AgNO_3$ using Pt electrodes.

$$\text{Anode: } H_2O_{(l)} \rightarrow \frac{1}{2}O_{2(g)} + 2H^+ \Rightarrow 2e^-$$

$$\text{Cathode: } 2Ag^+_{(aq)} + 2e^- \rightarrow 2Ag_{(s)}$$

$$2Ag^+ + H_2O_{(l)} \rightarrow 2Ag_{(s)} + \frac{1}{2}O_{2(g)} + 2H^+$$

$$(\text{Cathode})\ \ (\text{Anode})$$

O_2 gas is evolved at the anode.

(C) A diluted solution of H_2SO_4 using Pt electrodes.

$$\text{Anode}: H_2O_{(l)} \rightarrow \frac{1}{2}O_{2(g)} + 2H^+ + 2e^-$$

$$\text{Cathode}: \frac{2H^+ + 2e^- \rightarrow H_{2(g)}}{H_2O_{(l)} \rightarrow H_{2(g)} + \frac{1}{2}O_{2(g)}}$$

$$(\text{Cathode})\ \ (\text{Anode})$$

(D) A high concentration solution of H_2SO_4 + using Pt electrodes.

$$\text{Anode}: 2SO_4^{2-}{}_{(aq)} \rightarrow S_2O_8^{2-}{}_{(aq)} + 2e^-$$

$$\text{Cathode}: \frac{2H^+_{(aq)} + 2e^- \rightarrow H_{2(g)}}{2H^+_{(aq)} + 2SO_4^{2-}{}_{(aq)} \rightarrow H_{2(g)} + S_2O_8^{2-}{}_{(aq)}}$$

No, O_2 gas is evolved.

37. Option (1) is correct.

Homoleptic complexes are the complexes in which similar kind of ligands are present.

From the given complexes following are the homoleptic complex: $[FeO_4]^{2-}$,$[Fe(CN)_6]^{3-}$ and $[CoCl_4]^{2-}$.

The oxidation state of central metal ion in the above complexes are +6, +3 and +2 respectively.

(1) Fe: [Ar] $3d^6\ 4s^2$

Fe^{+6}: [Ar] $3d^2\ 4s^0$

Being a weak field ligand O^{2-}, does not pair with the unpaired e^- $\therefore$ an even no. of unpaired electrons are present in the $[FeO_4]^{2-}$ complex.

(2) Fe: [Ar]$3d^6\ 4s^2$

Fe^{+3}: [Ar] $3d^5\ 4s^0$

Being a strong field ligand, CN pairs with the unpaired e$-$ $\therefore$an odd no. of unpaired electrons are present in the $[Fe(CN)_6]^{3-}$ complex.

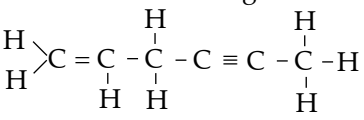
(3) Co: [Ar] $3d^7\ 4s^2$

Co^{+2}: [Ar] $3d^7$

Being a weak field ligand, Cl^-does not pair with the unpaired e^- $\therefore$an odd no. of unpaired electrons are present in the $[CoCl_4]^{2-}$ complex.

38. Option (1) is correct.

The structure of the given IUPAC name:



No. of sigma (σ) bonds is 13.

No. of pi (π) bonds is 3.

39. Option (4) is correct.

(1) $C_{(diamond)} + O_{2(g)} \rightarrow CO_{2(g)} + Y\ kJ\,mol^{-1}$

(2) $CO_{2(g)} \rightarrow C_{(graphite)} + O_{2(g)} - Z\ kJ\,mol^{-1}$

(3) $C_{(diamond)} \rightarrow C_{(graphite)} + X\ kJ\,mol^{-1}$

From equation (1), (2) and (3), $X = Y - Z$.

40. Option (2) is correct.

Statement (I) is correct: It is not possible to determine simultaneously, the position and momentum of a particle.

Statement (II) is correct: Given $\Delta p = \Delta x$,

from $\Delta p \times \Delta x \geq \frac{h}{4\pi}$

as $\Delta p = m\Delta v$ and $\Delta p = \Delta x$.

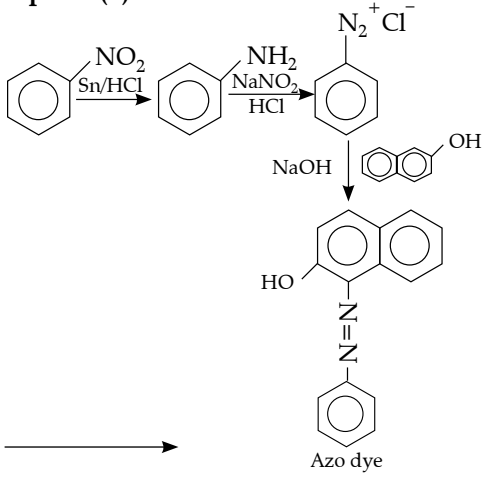
So $(\Delta p)^2 \geq \frac{h}{4\pi,}$

$(m\Delta v)^2 \geq \frac{h}{4\pi,}$

$\Delta v \geq \sqrt{\frac{h}{4\pi m^2,}}$

or $\Delta v \geq \frac{1}{2m} \times \sqrt{\frac{h}{\pi.}}$

41. Option (1) is correct.



42. Option (4) is correct.

Given initial concentration = 16 mg/mL,

final concentration = 4 mg/mL.

Time = 12 months.

$$\text{From } K = \frac{2.303}{t}\log\frac{a}{(a-x)} = \frac{2.303}{12}\log\frac{16}{4}$$

$$= \frac{2.303}{12}\log 4$$

$$= \frac{2.303 \times 0.6020}{12}$$

$$K = 0.1155\ \text{month}^{-1}$$

$$\text{From } t_{\frac{1}{2}} = \frac{0.693}{K} = \frac{0.693}{0.1155\ \text{month}^{-1}} = 6\,\text{month}$$

The expiry time of the drug in months is 6.

43. Option (1) is correct.

The given elements Li, Na, Be, Mg, B and Al are present in the periodic table as follows:

Period

2 Li Be B ↓ Top to bottom atomic radii increase

3 Na Mg Al

Left to right atomic radii decrease

The element with the least atomic radii is B, which forms B_2O_3 type oxide, so the correct answer is A_2O_3.

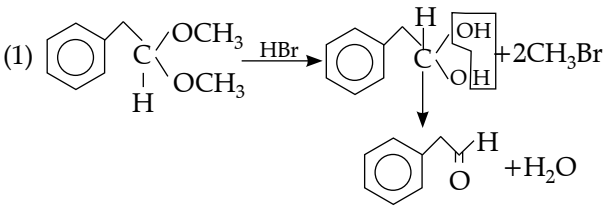
44. Option (1) is correct.

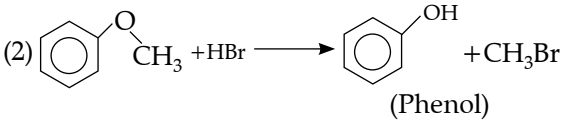
The element which is a main component of the apatite family is phosphorous atom.

Formula of the apatite family = $Ca_{10}(PO_4)_6 \cdot Cax_2$

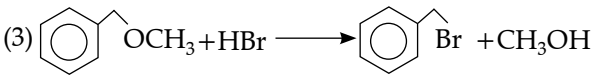
I.E ($Kjmol^{-1}$)	N	P	As	Sb
	1402	1012	947	834

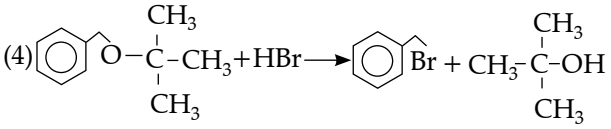
45. Option (2) is correct.





(Phenol)





46. Correct answer is [0].

The oxidation state of metal ion in the complex are:

Complex	Oxidation State	Strength of Ligand	No. of Unpaired e^-
$K_3[Co(No_2)_6]$	$Co^{+3}(d^6)$	Strong	0
$K_4[Fe(CN_6)]$	$Fe^{+2}(d^6)$	Strong	0
$K_3[Fe(CN_6)]$	$Fe^{+3}(d^5)$	Strong	1
$Cu_2[Fe(CN_6)]$	$Fe^{+2}(d^6)$	Strong	0
$Zn_2[Fe(CN_6)]$	$Fe^{+2}(d^6)$	Strong	0

In the above complex, only one complex contains the unpaired e^- and the colour of the complex is not yellow.

47. Correct answer is [2].

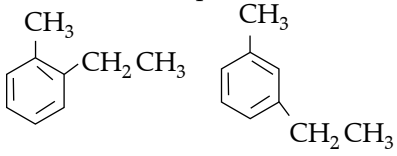
First we calculate the degree of unsaturation:

$$D.U. = \frac{C_n(v-2)}{2}+1 = \frac{9(4-2)+12(1-2)}{2}+1$$

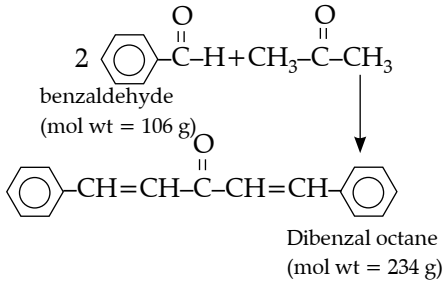
$$= \frac{18-12}{2}+1 = 4$$

For $D.U. = 4$ (Benzene ring).

The possible structure of C_9H_{12} with four different non aliphatic substitution sites is:



48. Correct answer in [60].



From the reaction it is clear that

2 × 106 gm Benzaldehyde produces 234 gm dibenzalacetone.

So 5.3 gm Benzaldehyde produces

$$\frac{234\times 5.3\text{gm}}{2\times 106} = 5.85\,\text{gm}.$$

$$\%\,\text{yield} = \frac{\text{actual wt}}{\text{theortical wt}}\times 100 = \frac{3.51\,\text{gm}}{5.85\,\text{gm}}\times 100$$

$$= 60\%.$$

49. Correct answer is [275].

Formula used,

$$\text{Percentage of sulphur} = \frac{\text{atomic mass of S}\times\text{mass of } BaSO_4\times 100}{\text{molor mass of } BaSO_4\times\text{weight of organic compound}}$$

Given weight of organic compound = 0.20 gm.

Mass of $BaSO_4$ produced = 0.40 gm.

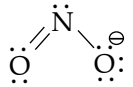
Molar mass of $BaSO_4$ = 233gm.

$$\%\,\text{of S} = \frac{32\times 0.40\,\text{gm}\times 100}{233\,\text{gm}\times 0.20} = 27.5\%$$

$$= 275\times 10^{-1}\%.$$

50. Correct answer is [12].

The structure of NO_2^- ion is:



The number of non-bonded election pair is 6.

So, the number of non-bonded electrons present in NO_2^- ion is = 6 × 2

=12.

Mathematics

Section A

51. Option (3) is correct.

Explanation: We have, $2x^2 + (a-5)x + 15 = 3a$

Has no real root $\Rightarrow D < 0$

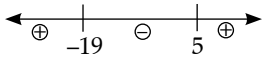
$\Rightarrow (a-5)^2 - 8(15-3a) < 0$

$\Rightarrow a^2 + 14a + 25 - 120 < 0$

$\Rightarrow a^2 + 14a - 95 < 0$

$\Rightarrow (a+19)(a-5) < 0$

$\Rightarrow a \in (-19, 5)$



$\therefore \sum_{x \in X} x^2 = \left(1^2 + 2^2 + \ldots + 4^2\right) + \left(1^2 + 2^2 + \ldots + 18^2\right)$

$= \frac{4 \times 5 \times 9}{6} + \frac{18 \times 19 \times 37}{6} = 30 + 2109 = 2139$

52. Option (3) is correct.

Explanation: We have, $\sin x + \sin^2 x = 1$

$\Rightarrow \sin x = 1 - \sin^2 x$

$\Rightarrow \sin x = \cos^2 x$ ...(i)

$\therefore$ Given expression:

$= 2\cos^{12} x + 6(\cos^{10} x + \cos^8 x) + 2\cos^6 x$

Put $\cos^2 x = \sin x$

$= 2\left[\sin^6 x + 3\sin^5 x + 3\sin^4 x + \sin^3 x\right]$

$= 2\sin^3 x\left[\left(\sin x + 1\right)^3\right]$

$[\because (a+b)^3 = a^3 + b^3 + 3a^2b + 3ab^2]$

$= 2\left[\sin^2 x + \sin x\right]^3 = 2$

53. Option (4) is correct.

Explanation: We have,

$c_1: |y| = 1 - x^2$ ...(i)

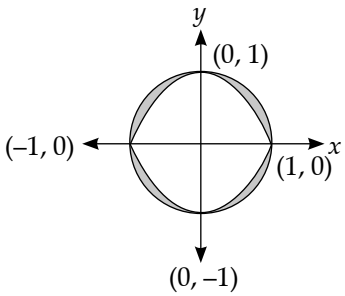
$c_2: x^2 + y^2 = 1$ ...(ii)

Solving (i) and (ii):

$1 - |y| + y^2 = 1$

$\therefore y = 0, 1, -1$

Circle: Centre (0, 0), Radius = 1



$\therefore$ Required area $= \alpha$

$= 4$ [area of circle in 1st quadrant $-\int_0^1 \left(1 - x^2\right) dx$]

$= 4\left[\frac{\pi}{4} - \left[x - \frac{x^3}{3}\right]_0^1\right]$

$\alpha = \pi - \frac{8}{3}$

$\therefore 3\alpha = 3\pi - 8$

$\Rightarrow 9\alpha = 9\pi - 24$

$\therefore \beta = 9, \gamma = -24$

$\therefore |\beta - \gamma| = |9 + 24| = 33$

54. Option (3) is correct.

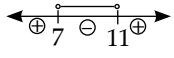
Explanation: Let, $f_1(x) = \log_5 (18x - x^2 - 77)$

$\therefore 18x - x^2 - 77 > 0$

$\because \log_b$ defines only, when $a > 0, b > 0, b \neq 1$

$x^2 - 18x + 77 < 0$

$(x-7)(x-11) < 0$



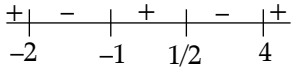
$x \in (7, 11)$

$\Rightarrow \alpha = 7, \beta = 11$

and $f_2(x) = \log_{(x-1)} \left(\frac{2x^2 + 3x - 2}{x^2 - 3x - 4}\right)$

$\therefore x - 1 > 0, x - 1 \neq 1, \frac{2x^2 + 3x - 2}{x^2 - 3x - 4} > 0$

$x > 1, x \neq 2, \ \frac{(2x-1)(x+2)}{(x-4)(x+1)} > 0$



and $x > 1$

$\therefore x \in [4, \infty)$

$\therefore \gamma = 4$

$\therefore \alpha^2 + \beta^2 + \gamma^2 = 49 + 121 + 16 = 186$

55. Option (1) is correct.

Explanation: $\because \cos|x|$ is always differentiable

$(\because \cos|x| = \cos x)$

Again, $|x^2 - ax + 2|$ is not differential when,

$x^2 - ax + 2 = 0$

Since, one root, $\alpha = 2$

$\therefore 4 - 2a + 2 = 0$

$\Rightarrow a = 3 \Rightarrow$ other root $\beta = 1$

The distance of $(\alpha, \beta) = (2, 1)$ from the line $12x + 5y + 10 = 0$, is

$d = \frac{12\alpha + 5\beta + 10}{\sqrt{12^2 + 5^2}} = \frac{12 \times 2 + 5 \times 1 + 10}{\sqrt{12^2 + 5^2}} = 3$

56. Option (3) is correct.

Explanation:

$L_1 : \frac{x-1}{2} = \frac{y+1}{1} = \frac{z-3}{-2} \Rightarrow \vec{b_1} = 2\hat{i} + \hat{j} - 2\hat{k}$

$L_2 : \frac{x-3}{1} = \frac{y-2}{3} = \frac{z+2}{4} \Rightarrow \vec{b_2} = \hat{i} + 3\hat{j} + 4\hat{k}$

$\because$ L is $\perp r$ to L_1 and L_2

Vector parallel to 'L'

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 3 & 4 \end{vmatrix} = 10\hat{i} - 10\hat{j} + 5\hat{k} \quad = 5(2\hat{i} - 2\hat{j} + \hat{k})$$

Equation of ' L'

$$\frac{x-2}{2} = \frac{y+1}{-2} = \frac{z-3}{1} = \lambda(\text{ say })$$

Let any point Q $(2\lambda + 2, -2\lambda - 1, \lambda + 3)$ on L

$\Rightarrow 2\lambda + 2 = 0$ ($\because$ Q lines on yz plane)

$\Rightarrow \lambda = -1 \Rightarrow$ Q (0, 1, 2)

$\therefore$ Perpendicular distance between P (2, –1, 3) and Q (0, 1, 2) is

$$= \sqrt{(2-0)^2 + (-1-1)^2 + (3-2)^2}$$

$$= \sqrt{4+4+1} = \sqrt{9} \ = 3$$

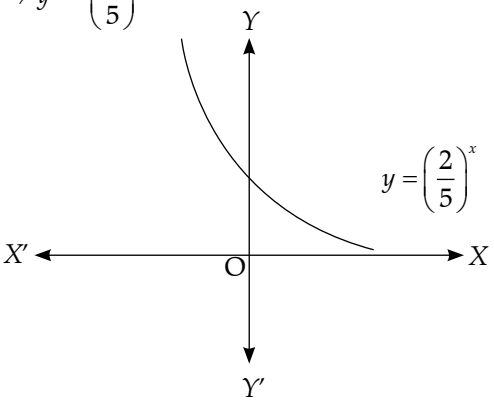
57. Option (2) is correct.

Explanation: S = [0, 1, 2, 3, ……]

$R = \{(x, y); \log_e y = x \log_e \frac{2}{5}, x \in S, y \in R\}$

$\because \log_e y = x \log_e \left(\frac{2}{5}\right) \ x \in S, y \in Ry$

$\Rightarrow y = \left(\frac{2}{5}\right)^x$



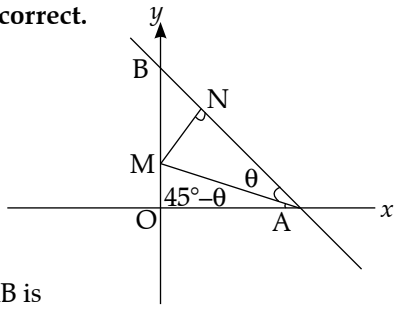
$\therefore$ Required sum $= 1 + \left(\frac{2}{5}\right)^1 + \left(\frac{2}{5}\right)^2 + \left(\frac{2}{5}\right)^3 + \ldots$

$$= \frac{1}{1 - \frac{2}{5}} = \frac{5}{3}$$

($\because$ Sum of infinite terms in GP is $\frac{a}{1-r}$)

58. Option (4) is correct.

Explanation:



$ar.(\text{AOB}) = \frac{1}{2}$

$ar\ (\Delta AMN)$

$= \frac{4}{9} \times \frac{1}{2} = \frac{2}{9}$

Equation of AB is

$x + y = 1$

OA = 1,

AM = sec (45° – θ)

AN = sec (45° – θ) cos θ

MN = sec (45° – θ) sin θ

Now, *ar* (ΔAMN)

$$= \frac{1}{2}\sec^2\left(45^\circ - \theta\right)\sin\theta \cdot \cos\theta = \frac{2}{9}$$

$\Rightarrow \tan\theta = 2, \frac{1}{2}$

$\tan\theta = 2$ is rejected

So, $\frac{AN}{NB} = \frac{\lambda}{1} = \cot\theta = 2$

59. Option (3) is correct.

Explanation: $E : \frac{x^2}{9} + \frac{y^2}{4} = 1$

Equation of chord $\Rightarrow \alpha x + \beta y = 109$ ...(i)

Equation of chord bisected at $\left(\frac{5}{2}, \frac{1}{2}\right)$

$T = S_1$

$$\frac{5}{2}\left(\frac{x}{9}\right) + \frac{1}{2}\left(\frac{y}{4}\right) = \frac{25}{36} + \frac{1}{16}$$

$$\Rightarrow \frac{5x}{18} + \frac{y}{8} = \frac{100+9}{144} = \frac{109}{144}$$

$\Rightarrow 40x + 18y = 109$ ...(ii)

$\Rightarrow \alpha = 40, \beta = 18$ (by comparing (i) and (ii))

$\Rightarrow \alpha + \beta = 58$

60. Option (3) is correct.

Explanation: We have,

A, K, N, P, R, U

The words starts with A

A ……. = 5! = 120

The words starts with K

K ……. = 5! = 120

The words starts with N

N …… = 5! = 120

The words starts with PA

P A …… = 4! = 24

The words starts with PK

P K …… = 4! = 24

The words starts with PN

P N …… = 4! = 24

The words starts with PRA

P R A …… = 3! = 6

P R K A N U …… = 1

P R K A U N …… = 1

Total no. of words = 440

$\Rightarrow \therefore 440^{th}$ word = P R K A U N

61. Option (1) is correct.

Explanation: Given: $x + y + z = 1$

$x + 2y + 4z = m$

$x + 4y + 10z = m^2$

Has infinite many solutions

$\Delta = \Delta x = \Delta y = \Delta z = 0$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{vmatrix}$$

$= 1(20 - 16) - 1(10 - 4) + 1(4 - 2) = 4 - 6 + 2 = 0$

For infinite solutions

$\Delta x = \Delta y = \Delta z = 0$

$m^2 - 3m + 2 = 0$

$m = 1, 2$

$\alpha = 1, \beta = 2$

$$\therefore \sum_{n=1}^{10}\left(n^{\alpha} + n^{\beta}\right) = \sum_{n=1}^{10} n^1 + \sum_{n=1}^{10} n^2$$

$$= \frac{10(11)}{2} + \frac{10(11)(21)}{6}$$

$= 55 + 385 = 440$

62. Option (4) is correct.

Explanation: $\therefore$ We have, A = [a_{ij}], 3 × 3 matrix

Where, $a_{ij} = (\sqrt{2})^i + j$

$$\Rightarrow A = \begin{bmatrix} \left(\sqrt{2}\right)^2 \left(\sqrt{2}\right)^3 \left(\sqrt{2}\right)^4 \\ \left(\sqrt{2}\right)^3 \left(\sqrt{2}\right)^4 \left(\sqrt{2}\right)^5 \\ \left(\sqrt{2}\right)^4 \left(\sqrt{2}\right)^5 \left(\sqrt{2}\right)^6 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 2 & 2\sqrt{2} & 4 \\ 2\sqrt{2} & 4 & 4\sqrt{2} \\ 4 & 4\sqrt{2} & 8 \end{bmatrix}$$

$$A^2 = 2^2 \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 2 & 2\sqrt{2} \\ 2 & 2\sqrt{2} & 4 \end{bmatrix} \times \begin{bmatrix} 1 & \sqrt{2} & 2 \\ \sqrt{2} & 2 & 2\sqrt{2} \\ 2 & 2\sqrt{2} & 4 \end{bmatrix}$$

Sum of elements of 3rd row,

$= 4(14 + 14\sqrt{2} + 28) = 4(42 + 14\sqrt{2}) = 168 + 56\sqrt{2}$

$= \alpha + \beta\sqrt{2}$

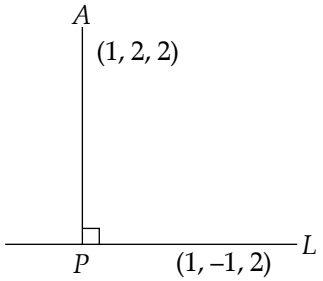
$\therefore \alpha + \beta = 168 + 56 = 224$

63. Option (1) is correct.

Explanation: $L : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2}$

P – foot of $\perp r$ drawn from A(1, 2, 2) on line L.

$\vec{b} = \hat{i} - \hat{j} + \hat{k}$



(1, – 1, + 2) = $\vec{d}$

$L : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{2} = \mu$(say)

Put $P(\mu + 1, -\mu - 1, 2\mu + 2)$

$\overrightarrow{AP}.\vec{d} = 0$

$\Rightarrow (\mu + 1 - 1) + (-\mu - 1 - 2)(-1) + 2(2\mu + 2 - 2) = 0$

$\Rightarrow \mu + \mu + 3 + 4\mu = 0$

$\Rightarrow \mu = \frac{-1}{2}$

$$\therefore P\left(\frac{-1}{2} + 1, +\frac{1}{2} - 1, 2\left(\frac{-1}{2}\right) + 2\right)$$

$$P\left(\frac{1}{2}, -\frac{1}{2}, 1\right)$$

$$L_2 : \frac{x+1}{1} = \frac{y-1}{-1} = \frac{z+2}{1} = \lambda$$

Now, general point on L_2 is

Q (– 1 + λ, 1 – λ, – 2 + λ)

Equate it with general point of L

$\mu + 1 = -1 + \lambda$	$-\mu - 1 = 1 - \lambda$
$\mu = \lambda - 2$	$\mu = \lambda - 2$

$2\mu + 2 = -2 + \lambda$

$2(\lambda - 2) + 2 = -2 + \lambda$

$\therefore \mu = -2, \lambda = 0$

$\Rightarrow$Q (–1, 1, – 2)

So, $PQ = \sqrt{\left(\frac{1}{2} + 1\right)^2 + \left(\frac{-1}{2} - 1\right)^2 + (1+2)^2}$

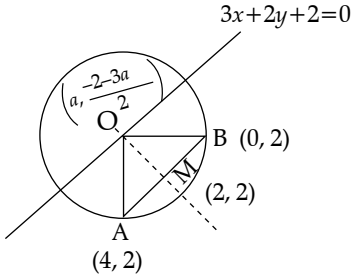
$$= \sqrt{\frac{9}{4} + \frac{9}{4} + 9} = \sqrt{\frac{54}{4}}$$

$$\therefore 2(PQ)^2 = 2\left(\frac{54}{4}\right) = 27$$

64. Option (2) is correct.

Explanation: Let Centre O $\left(a, \frac{-2-3a}{2}\right)$

Mid point AB = (2, 2) → M



Now

$$\text{(Slope of OM)}\left(\frac{-3}{2}\right)=-1\Rightarrow\frac{2+\frac{2+3a}{2}}{2-a}\left(\frac{-3}{2}\right)=-1$$

$\Rightarrow (6 + 3a)(-3) = -4(2 - a)$

$\Rightarrow -18 - 9a = -8 + 4a$

$\Rightarrow 13a = 2b \Rightarrow a = 2$

$\therefore$ Centre O = $(2, -4)$

$$r = OA = \sqrt{(4-2)^2+(2+4)^2} = \sqrt{40}$$

Midpoint of chord is N = (1, 2)

$\therefore$ ON = $\sqrt{37}$

$\therefore$ Length of chord = $2\sqrt{r^2-(OM)^2}$

$= 2\sqrt{40-37} = 2\sqrt{3}$

65. Option (2) is correct.

Explanation: We have, $A=\begin{bmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix}$

$|A| = a_{11}\ a_{22} - a_{12}\ a_{21} = \{-1, 0, 1\}$

x_i	P_i	$P_i x_i$	$P_i x_i^2$
– 1	$\frac{3}{16}$	$-\frac{3}{16}$	$\frac{3}{16}$
0	$\frac{10}{16}$	0	0
1	$\frac{3}{16}$	$\frac{3}{16}$	$\frac{3}{16}$

$$\sum p_i x_i = 0 \quad i\sum p_i x_i^{\,2} = \frac{3}{8}$$

$\therefore$ Var(x) = $P_i X_i^2 - (P_i X_i)^2 = \frac{3}{8} - 0 = \frac{3}{8}$

66. Option (4) is correct.

Bag 1 = [4W, 5B]

Bag 2 = [nW, 3B]

$$p\left(\frac{W}{Bag_2}\right)=\frac{29}{45}$$

According to the question,

$$\Rightarrow\ p\left(\frac{W}{B_1}\right)\times p\left(\frac{W}{B_2}\right)+p\left(\frac{B}{B_1}\right)\times p\left(\frac{W}{B_2}\right)=\frac{29}{45}$$

$$\Rightarrow \frac{4}{9}\times\frac{n+1}{n+4}+\frac{5}{9}\times\frac{n}{n+4}=\frac{29}{45}\Rightarrow n=6$$

67. Option (1) is correct.

Explanation:

$7^{103} = 7(7)^{102} = 7(343)^{34} = 7(345-2)^{34}$

$= 7({}^{34}C_0(345)^{34}(2)^0 - {}^{34}C_1(345)^{33}2 + \ldots {}^{34}C_4 2^{34})$

$\Rightarrow 7^{103} = 23K_1 + 7.2^{34}$

Now, $7.2^{34} = 7.2^2.2^{32}$

$= 28.(256)^4 = 28(253 + 3)^4$

$\therefore 28 \times 81 \Rightarrow (23 + 5)(69 + 12) = 23K_2 + 60$

$\therefore$ Remainder = 14

68. Option (1) is correct.

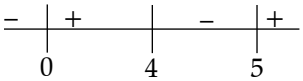
Explanation: We have,

$$f(x)=\int_0^x t(t^2-9t+20)\,dt, 1\le x\le 5$$

Using Leibnitz Rule,

$f'(x) = x^3 - 9x^2 + 20x$ ...(i)

$= x(x-4)(x-5)$



$\therefore$ Function is increasing in x (0, 4) $\cup$ (5, ∞)

Integrating equation (i)

$$\therefore\quad f(x)=\frac{x^4}{4}-\frac{9x^3}{3}+\frac{20x^2}{2}+c$$

$\because f(0) = 0 \Rightarrow c = 0$

Now, $f(1)=\frac{1}{4}-3+10=\frac{29}{4}=\alpha$

$$f(4)=\frac{256}{4}-3(64)+10\times(16)\ =32$$

$$\text{Range}=\left(\frac{29}{4},32\right],\ \alpha=\frac{29}{4},\beta=32$$

$$\Rightarrow 4(\alpha+\beta)=4\left(\frac{29}{4}+32\right)=157$$

69. Option (3) is correct.

Explanation: Let $\vec{P}=\hat{i}+\hat{j}+\hat{k}$

$$\vec{b}\times\vec{c}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -2 & 3\\ 2 & 3 & -1\end{vmatrix}=-7\hat{i}+7\hat{j}+7\hat{k}$$

$= -7(\hat{i}-\hat{j}-\hat{k})\ \left(\because \vec{a}\text{ is}\perp r\text{ to}\vec{b}\text{ and}\vec{c}\right)$

Now, $\hat{a}=\frac{\hat{i}-\hat{j}-\hat{k}}{\sqrt{3}}$ or $\frac{-\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}$

$$\cos\theta=\frac{\hat{a}\cdot\vec{p}}{|\vec{p}|}=\frac{1-1-1}{\left(\sqrt{3}\right)^2}$$

$$\text{or }\cos\theta=\frac{-1+1+1}{3}=\frac{1}{3}$$

$$\vec{a} = \frac{\hat{i}-\hat{j}-\hat{k}}{\sqrt{3}}$$

Now, $\cos\frac{\pi}{3} = \left(\frac{\hat{i}-\hat{j}-\hat{k}}{\sqrt{3}}\right)\frac{(\hat{i}+\alpha\hat{j}+\hat{k})}{\sqrt{1+\alpha^2+1}}$

$\Rightarrow \frac{1}{2} = \frac{1-\alpha-1}{\sqrt{3}\sqrt{\alpha^2+2}} \Rightarrow \frac{\sqrt{3}}{2}\sqrt{\alpha^2+2} = -\alpha$

Squaring both sides, we get

$\Rightarrow 3\alpha^2 + 6 = 4\alpha^2$

$\Rightarrow \alpha = \pm\sqrt{6}$

$\because \alpha < 0$

$\alpha = -\sqrt{6}$

70. Option (4) is correct.

Explanation: $\frac{dy}{dx} + (\tan x)y = \frac{2+\sec x}{(1+2\sec x)^2}$

It is (LDE)

$P = \tan x$ and, $Q = \frac{2+\sec x}{(1+2\sec x)^2}$

IF $= e^{\int \tan x dx} = e^{\ln(\sec x)} = \sec x$

$$\therefore y\cdot\sec x = \int\left\{\frac{2+\sec x}{(1+2\sec x)^2}\right\}\sec x\,dx \qquad ...(i)$$

Let $I = \int \frac{2\cos x+1}{(\cos x+2)^2}dx$

Let $\cos x = \frac{1-t^2}{1+t^2}$

$$= \int \frac{2\left(\frac{1-t^2}{1+t^2}\right)+1}{\left(\frac{1-t^2}{1+t^2}+2\right)^2}\times\frac{2\,dt}{(1+t^2)}$$

$$= \int \frac{2-2t^2+1+t^2}{\left(1-t^2+2+2t^2\right)^2}\times 2dt = 2\int\frac{3-t^2}{\left(t^2+3\right)^2}dt$$

Let $t + \frac{3}{t} = u$

$$\left(1-\frac{3}{t^2}\right)dt = du = -2\int\frac{du}{u^2} = \frac{2}{u}+c$$

Solution of DE

$$y\cdot(\sec x) = \frac{2}{u}+c$$

$$y\cdot\sec x = \frac{2}{t+\frac{3}{t}}+c \qquad ...(i)$$

At $x = \frac{\pi}{3}, t = \tan\frac{x}{2} = \frac{1}{\sqrt{3}}$

$$2\cdot\frac{\sqrt{3}}{10} = \frac{2}{\frac{1}{\sqrt{3}}+3\sqrt{3}}+c$$

$$\frac{2\sqrt{3}}{10} = \frac{2\sqrt{3}}{10}+c$$

$\Rightarrow \quad c = 0$

at $x = \frac{\pi}{4}, t = \tan\frac{x}{2} = \sqrt{2}-1$

$$\therefore \quad y\cdot\sqrt{2} = \frac{2}{\sqrt{2}-1+\frac{3}{\sqrt{2}-1}}$$

$$y\cdot\sqrt{2} = \frac{\left(2-\sqrt{2}\right)}{6-2\sqrt{2}}$$

$$y = \frac{\left(2-\sqrt{2}\right)}{2(3-\sqrt{2})}$$

$$= \frac{1}{2}\times\frac{\left(4-\sqrt{2}\right)}{7} = \frac{4-\sqrt{2}}{14}$$

Section B

71. Correct answer is [12].

Explanation:

$$\because \quad 24\left[\int_0^{\frac{\pi}{48}} -\sin\left(4x-\frac{\pi}{12}\right)+\int_{\frac{\pi}{48}}^{\frac{\pi}{4}}\sin\left(4x-\frac{\pi}{12}\right) + \int_0^{\frac{\pi}{6}}[0]\,dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}}[2\sin x]dx\right]$$

$$= 24\left[\frac{\left(1-\cos\frac{\pi}{12}\right)}{4} - \frac{\left(-\cos\frac{\pi}{12}-1\right)}{4}\right] + \frac{\pi}{4} - \frac{\pi}{6}$$

$$= 24\left(\frac{1}{2}+\frac{\pi}{12}\right) = 2\pi+12$$

$\Rightarrow \alpha = 12$

72. Correct answer is [64].

Explanation:

$$\lim_{t\to 0}\left[\int_0^1 (3x+5)^t\,dx\right]^{\frac{1}{t}}$$

$$= \lim_{t\to 0}\left\{\left[\frac{(3x+5)^{t+1}}{3(t+1)}\right]_0^1\right\}^{\frac{1}{t}} = \lim_{t\to 0}\left[\frac{8^{t+1}-5^{t+1}}{3(t+1)}\right]^{\frac{1}{t}}$$

$t = 0\ 1^\infty$ form

$$e^{\lim_{t\to 0}\left[\frac{8^{t+1}-5^{t+1}-3(t+1)}{3(t+1)}\right]\cdot\frac{1}{t}}$$

Using L Hospital's rule

$$\lim_{t\to 0} e^{\left[\frac{\left(8^{t+1}\right)\ln 8-\left(5^{t+1}\right)\ln 5-3}{3(2t+1)}\right]}$$

Put $t = 0$

$$= e^{\frac{8\ln 8-5\ln 5-3}{3}}$$

$$= \left(\frac{8}{5}\right)^{\frac{2}{3}}\left(\frac{64}{5e}\right) = \frac{\alpha}{5e}\left(\frac{8}{5}\right)^{\frac{2}{3}} \text{ (given)}$$

On comparing, we get

$\alpha = 64$

73. Correct answer is [11132].

Explanation: Given,

$a_1 + a_5 + a_{10} + \ldots.. + a_{2020} + a_{2024} = 2233$

In the AP, the sum of terms equidistant from ends are equal.

$a_1 + a_{2024} = a_5 + a_{2020} = a_{10} + a_{2015}$

$= 203$ pairs

$= 203\ (a_1 + a_{2024}) = 2233$

Hence, $S_{2024} = \frac{2024}{2}\left(a_1 + a_{2024}\right)$

$$= 1012 \times 11 = 11132$$

74. Correct answer is [10].

Explanation: We have,

$a, b, \in I, -3 \le a, b \le 3$

$a + b \neq 0$

$\Rightarrow z|z-a| = |z+b|$

$$\Rightarrow z\begin{vmatrix} z+1 & (\omega) & (\omega)^2 \\ (\omega) & z+(\omega)^2 & 1 \\ (\omega)^2 & 1 & z+(\omega) \end{vmatrix} = 1$$

$$\Rightarrow z\begin{vmatrix} z & z & z \\ (\omega) & z+(\omega)^2 & 1 \\ (\omega)^2 & 1 & z+(\omega) \end{vmatrix} = 1$$

$$\Rightarrow z\begin{vmatrix} 1 & 1 & 1 \\ (\omega) & z+(\omega)^2 & 1 \\ (\omega)^2 & 1 & z+(\omega) \end{vmatrix} = 1$$

$$\Rightarrow z\begin{vmatrix} 1 & 0 & 0 \\ (\omega) & z+(\omega)^2-(\omega) & 1-(1) \\ (\omega)^2 & 1-(\omega)^2 & z+(\omega)+(\omega)^2 \end{vmatrix} = 1$$

$\Rightarrow z^3 = 1 \Rightarrow z = \omega, \omega^2, 1$

Now, $|1-a| = |1+b|$

$\Rightarrow$ So, total number of pairs = 10

75. Correct answer is [1328].

Explanation: We have, $y^2 = 12x$

$\because 4a = 12$

$\Rightarrow a = 3$

And $SP \times SQ = \frac{147}{4}$

Let $P\ (3t^2, 6t)$ and $t_1t_2 = -1$

So, $Q\left(\frac{3}{t^2}, \frac{-6}{t}\right)$

Now, focus S $(a, 0)$

$\Rightarrow$ S $(3, 0)$

$\therefore\ SP \times SQ = PM_1 \times QM_2$

(distance from directrix)

$$\left(3+3t^2\right)\left(3+\frac{3}{t^2}\right) = \frac{147}{4}$$

$$\Rightarrow \frac{\left(1+t^2\right)^2}{t^2} = \frac{49}{12}$$

$$\Rightarrow t^2 = \frac{3}{4}, \frac{4}{3}$$

$$\Rightarrow t = \pm\frac{\sqrt{3}}{2}, \pm\frac{2}{\sqrt{3}}$$

Considering $t = -\frac{\sqrt{3}}{2}$

$P\left(\frac{9}{4}, -3\sqrt{3}\right)$ and $Q(4, 4\sqrt{3})$

Hence, the diametric circle

$$(x-4)\left(x-\frac{9}{4}\right)+(y+3\sqrt{3})(y-4\sqrt{3}) = 0$$

$$\Rightarrow x^2 + y^2 - \frac{25}{4}x - \sqrt{3}y - 27 = 0$$

$\Rightarrow \alpha = 400, \beta = 1728$

$\Rightarrow \beta - \alpha = 1328$

■■■